One Machine: the unified edition

OES end-to-end model, 28 October 2026. This edition puts the whole model in one text with one ledger. It brings together the verified build (layers 1–18), the integrated spine (layer 19), the canon synchronization (layer 20), the sibling's 28 October ledger of joins (layer 21), and the sibling's evening notes with the canon's 19:10 continuation (layer 22). It is a selected mathematical build replayed from matrices. It is not a source admission and not a change to the canon matrices.

AUTH = NOT_PASS · ScientificGold = FALSE · TIME_BIND = UNBOUND · DYN_BIND = NOT_INSTALLED · SCR6 = NOT_RUN_OPERAND_ABSENT · PredictionFreeze = NOT_TRIGGERED. No gate moves, and no term is added to the action.

Custody.

- Repository `sachinkaria24-byte/OES`, branch `claude/lagrangian-spectral-spacetime-model-t2253b`.
- `python verify.py` replays 22 layers in **462 checks**. Of these, 422 hold: THEOREM 299, OBSTRUCTION 41, ARITHMETIC 37, CONDITIONAL 17, UNFORMED 18, SELECTED_REFERENCE 8, REMOVED 2. The other 40 are REPORTED records.
- `python -m pytest` runs 39 tests.
- `ports/ts/` holds three drop-in audits for the sibling's app (41 rows, residual 0). `data/clocks.json` holds the explicit clock matrices, and `docs/LEDGER.md` holds the unified ledger.
- This text supersedes `one-machine-20261028-verified.pdf` (layers 1–18) and `one-machine-20261028-spine.pdf` (layers 19–20) as the current paper. Both stay in `docs/paper/` as history.

Abstract

The model has one datum, the signless incidence of K5. It has one action, fifteen oscillators with stiffness D^2, and one bath, the E8 root lattice. Everything else is a map between these three objects, a count, or a gate. This edition does four things.

It collapses the ledger. The sibling's 28 October ledger states about forty joins. The 129 sentences of the form "X contains every tick" or "X misses every nonzero tick" all carry one bit: the tick 864 lies in ⟨X⟩ exactly when 5 does not divide X. All 129 are checked against that one line, together with their step counts, their indices and 38 stated subgroup orders. The numbers that ride on the bit are shared integers, and no map joins them.

It puts every clock in one group. The normalizer of the Coxeter clock in W(E8) has order 240. In it:

- the tick is the commutator of the day clock and the Coxeter clock: $d\, c'\, d^{-1}\, c'^{-1} = c'^{6} = r\, r'$, two carrier rotations at once;
- the day–Coxeter group has order 120, centre $\langle c'^{5} \rangle$ and derived group the tick. It permutes its five day clocks as the affine group of the pentagon: the tick rotates them and the day doubles them, pentagon to pentagram;
- the golden clock has order 20, and its fourth power is the tick;
- the circle's mirror $x \mapsto -x$ is an involution that acts on the carrier as the pentagon reflection, fixing three atoms;
- the carrier and its twin can be exchanged only by elements of order 4, 8, 12 or 20, never by an involution, and every exchange doubles their index-5 glue.

The order-24 day element, which the sibling's build reports as not constructed, is given explicitly.

It states what is physical and what is not.

- The energy splits into four conserved pieces, and the vertex and edge sectors never exchange energy.
- The 240 roots are D8 plus one half-spin orbit.
- The source oscillator, of frequency 2, is not a carrier mode.
- Compton and Bohr lengths, the photoelectric effect, the double slit and entanglement stay unformed. No operand is stored for them, and the action has no dimensionful constant.

It replays the evening notes (layer 22).

- The sibling's 28 notes of 18:32–19:21 add 223 readings on the bit, and every one holds.
- Seven moments that the sibling reported as not replayed are the moments of the alternating hexagon port on the E8 lattice. They are computed exactly from the 240 roots, and the eighth is new.
- The canon's 19:10 continuation asks two things. A commutator should not be called 864 without a map: every map from the day–Coxeter group to the circle kills it. Equal energy totals do not make equal splits: the three splits are the margins of one conserved table.

0. The model on one page

object	definition	what is proved	layer
datum	signless incidence M of K_n ($n \times C(n,2)$)	dim ker M = n holds only at n = 5	1
carrier	$D = [[0, M], [M^T, 0]]$ on $\ell^2(V \sqcup E)$, dim 15	spec $D^2 = \{0^5, 3^8, 8^2\}$, Tr $D^2 = 40$, Tr $D^4 = 200$	1
action	$S = \int \tfrac{1}{2}\lvert\dot\psi\rvert^2 - \tfrac{1}{2}\langle\psi, D^2\psi\rangle$	15 oscillators: 5 free, 8 at √3, 2 at 2√2; four conserved energies	1, 21
spectral triple	$(\mathbb{C}^V \oplus \mathbb{C}^E, \ell^2(V \sqcup E), D, \gamma)$	heat trace $5 + 8e^{-3\beta} + 2e^{-8\beta}$; Witten index −5	2, 5
observer	Kalman readouts of $(\psi, \dot\psi)$	dark sector P5 ⊕ P5; five channels complete it	5, 20
bath	E8 root lattice, Δ = 240 − A	band [0, 256] (256 attained; the upper bound reported); 11 critical families, 243,600 points, Morse sum 0; group speed √30	3, 8, 16, 20
walks	W_n = closed walks of length n	one shift operator of order 16 and degree 58; its 11 in-band roots are the 11 critical values	14, 16
native action	source q on P5 ⊕ P5, quadratic port W_AS, six-site clique	$\ddot q + 4q = 0$ at the origin; sharp quartic 40/9	4, 10
glue	E8 ⊃ A4 ⊕ A4′	index 5, glue $x \mapsto \pm 2x$; no involution exchanges the two	11, 21
waves	Hodge complex on the K5 edges	sound (4) and light (6) are isochronous at √5; Agni = Curl^2	17
circle	$Z/4320 = 2^5 \cdot 3^3 \cdot 5$	one lemma; the tower 36 × 4 × 3 × 2 × 5; the tick bit	19, 21
clocks	the normalizer of ⟨c′⟩ in W(E8), order 240	every circle unit realized; the tick = [day, Coxeter]	19, 21

object	definition	what is proved	layer
curvature, cosmology	Lin–Lu–Yau; spectral action on $M^4 \times F$	discretely flat bath; G and Λ_{cc} only with external inputs (CONDITIONAL)	6, 7
gates	AUTH, TIME_BIND, DYN_BIND, PredictionFreeze, W_L, T3081	none moves	13

1. The datum, the action and the energy

The datum. For the complete graph K_n, the signless incidence M (vertex × edge) has dim ker M = n exactly when n = 5, for every n. This is the only input. The fifteen atoms are the 5 vertices and the 10 edges of K5.

The carrier and its action.

- $D = [[0, M], [M^T, 0]]$ is odd, with grading $\gamma = +1$ on vertices and -1 on edges.
- $D^2 = \mathrm{diag}(MM^T, M^TM)$, with $MM^T = 3I + J$ on the vertices. The spectrum is $\{0^5, 3^8, 8^2\}$, so $\mathrm{Tr}\, D^2 = 40$ and $\mathrm{Tr}\, D^4 = 200$. The kernel $\ker D = P5 \subset \ell^2(E)$ is the irreducible 5 of S5.
- The action $S = \int \tfrac{1}{2}|\dot\psi|^2 - \tfrac{1}{2}\langle\psi, D^2\psi\rangle$ has no other term. A relative constant between its two terms is a rescaling of t.
- Its Euler–Lagrange equation $\ddot\psi + D^2\psi = 0$ is solved in closed form sector by sector. The frequencies are 0, $\sqrt{3}$ and $2\sqrt{2}$, and their squared ratio 8/3 is not a rational square.

Four conserved energies (THEOREM, layers 5 and 21). D^2 is block diagonal in $V \oplus E$, so the vertex and edge sectors never exchange energy. Each spectral sector $\lambda \in \{0, 3, 8\}$ is also invariant. Together these give four conserved pieces.

The sibling's checked datum takes:

- vertex value $(1, -1, 0, 0, 0) + (1, 1, 1, 1, 1)$;
- edge value M^T applied to the contrast;
- velocity the alternating cycle on the edges 01, 12, 23, 03.

Its quadratic form is 64 = 40 + 6 + 18, and its energy is

> 34 = 20 (uniform vertex mode) + 3 (vertex contrast) + 9 (edge image of the contrast) + 2 (the cycle).

The vertex energy E_V = 23 and the edge energy E_E = 11 stay constant along the exact motion. The ledger's "34 = 12 + 20 + 2" is the spectral grouping of the same four pieces, with 12 = 3 + 9.

Decoupling, quantized (CONDITIONAL). Under canonical quantization of S, the oscillating modes evolve by $U_V(t) \otimes U_E(t)$. The carrier therefore never entangles vertices with edges; entanglement can only be present in the initial data. The ledger's "entanglement is not stored" becomes a sharper statement: the carrier cannot create it. The quantization is a choice the canon does not store.

Symmetries of the action (THEOREM).

- The graph reflection of the pentagon, $i \mapsto -i$ on the vertices, keeps D^2 and fixes three atoms (one vertex and two edges).
- The fifteen-atom involution $i \mapsto 14 - i$ (the ledger's "above and below") fixes one atom and has eigenspaces of ranks 8 and 7. It sends a vertex to an edge, so it does not keep D^2.

The harmonic quartic (THEOREM). The S5-invariant quartics on the contrast plane are spanned by $|x|^4$ and Σx_i^4. For the plane's own Laplacian, the harmonic line is $2|x|^4 - 5\Sigma x_i^4$. For the ambient Laplacian of $\mathbb{R}^5$, it is $3|x|^4 - 7\Sigma x_i^4$. Neither enters the action: coupling a quartic would add a dimensionful constant.

The source is not a carrier mode (OBSTRUCTION). The stored source equation is $\ddot{q} + 4q = 0$, with frequency 2. Since $4 \notin \text{spec } D^2 = \{0, 3, 8\}$, the source oscillator is not one of the fifteen carrier modes.

2. The spectral triple and the observer

Spectral content (THEOREM, layer 2).

- The heat trace is $Z(\beta) = 5 + 8e^{-3\beta} + 2e^{-8\beta}$.
- The moments are $\text{Tr } D^{2k} = 15, 40, 200, 1240, 8840$.
- The spectral zeta function is $\zeta_D(s) = 8 \cdot 3^{-s} + 2 \cdot 8^{-s}$.
- The spectral dimension tends to 0 in both the UV and the IR. The carrier has no emergent continuum dimension.

Observation (THEOREM, layers 5 and 20).

- A scalar readout of $x = (\psi, \dot{\psi})$ sees at most 6 of the 30 dimensions, two for each spectral line.
- The Witten index is $\text{Tr } \gamma\, e^{-\beta D^2} = -5$.
- The dark sector, reachable by edge forcing but invisible to $D\psi$, is the native source space $P5 \oplus P5$ (10 dimensions).

Five fixed channels are necessary and sufficient to complete the observer. With $H = [D; L]$, the reconstruction R satisfies $RH = I_{15}$, and $P_0 = I - 11K/24 + K^2/24$ with $K = D^2$. Of the 252 five-edge sensor designs, 162 are complete and 90 have rank 4. Every complete design has determinant 1/162. The 72 optimal designs have minimum eigenvalue $(3 - \sqrt{5})/6$.

3. The bath

Roots (THEOREM, layers 3 and 21).

- The 240 roots of E8 are 112 integer roots, which form D8, and 128 half-integer roots, which form one W(D8)-orbit (a half-spin weight set). So $248 = so(16) + 128$.
- The root frame is $\Sigma\alpha\alpha^T = 60I_8$, and the root graph is 56-regular.
- $[W(E8) : W(D8)] = 135$. This is the D8 family at the top of the band, and the half-turn's 256 fixed momenta are $1 + 120 + 135$.
- The ledger's "SU(2) is not derived" stands.

The band and its critical points (THEOREM, layers 3, 8, 14, 16). The symbol is $\varepsilon(k) = 240 - \Sigma \cos(\alpha \cdot k)$. It attains 0 and 256, the top at the band vertex $\pi(1,1,1,1,0,0,0,0)$. That the range is exactly [0, 256] is reported (an alcove branch-and-bound, layer 8), and the census below agrees with it. The census covers the vertices and edges of the alcove and lists $\lambda = 240 - \varepsilon$, the number of points and the Morse index:

family	λ	points	index		family	λ	points	index
E8	240	1	0		A6+A1 (edge)	−185/16	69,120	6
E7+A1	16	120	1		A7 (edge)	−320/27	17,280	7
E6+A2	−3	2,240	2		A7+A1	−12	8,640	7

family	λ	points	index		family	λ	points	index
D5+A3	−8	15,120	3		A8	−12	1,920	8
A4+A4	−10	48,384	4		D8	−16	135	8
A5+A2+A1	−11	80,640	5					

- There are 243,600 points in all, and the Morse sum is $0 = \chi(T^8)$.
- The eleven values of λ are exactly the eleven in-band roots of the walk operator below, so the band geometry and the walk data are one statement.

The walks (THEOREM, layer 14).

- $W_0 \ldots W_{90}$ are exact from two independent engines, and $W_0 \ldots W_{600}$ are known modulo primes.
- Every recurrence with $(r+1)(d+2) \le 455$ is excluded.
- The annihilator is one operator L_{16}, of order 16 and degree 58, the same modulo four primes. Its characteristic polynomial has the eleven in-band roots, $(z+12)^2$ among them, and five roots outside the band.
- Asymptotically, $W_n = (16/\pi^4)\cdot 240^n \cdot n^{-4} \cdot (1 - 2/n + (23/15)/n^2 + O(n^{-3}))$.

Speed and density.

- The long-wave dispersion is $\varepsilon = 30|k|^2 - (3/2)|k|^4 + (1/24)|k|^6 + \ldots$, so the sound speed is $\sqrt{30}$ (layer 3). This is also the bath's maximal group speed, as a supremum that is never attained: $|\nabla\varepsilon|^2 \le 120\varepsilon$, by the frame and Cauchy–Schwarz (layer 20).
- The nine positive critical energies have distinct square-free parts, so their frequencies are independent over $\mathbb{Q}$ and their flow is dense on T^9 (layer 18).

Open. The convexity collar (223.58, 224) remains open.

4. The carrier inside the bath

Native action (layers 4 and 10).

- The source $q = (a, b)$ lives in $P5 \oplus P5$, with mass $M_s = I/4$.
- The port $g = W_{AS}(a \wedge b)$ is quadratic onto the six-dimensional cycle space L6, with $W_{AS}^T W_{AS} = (10/9)P$ and sharp quartic constant 40/9.
- The six-site clique has $A_{bb} = 241I - J$ and exterior coupling $BB^* = 183P_c + 495P_u$.
- At linear order the source obeys $\ddot{q} + 4q = 0$ and has no linear channel into the bath. Its frequency-4 harmonic lies inside the band through a thin channel.
- The printed chart T_6 preserves the contrast lattice, and $T_6|V_c$ has determinant 31, so $155 = 5 \times 31$.

Glue (THEOREM, layers 11 and 21). The carrier's own root system $A4 = \{e_i - e_j : i, j \le 5\}$ sits in E8, and its orthogonal complement is a twin A4′. The index is $[E8 : A4 \oplus A4'] = 5$, and the glue is $x \mapsto \pm 2x$. Layer 21 completes the picture with the stabilizer of the pair {A4, A4′} in W(E8), which has 57,600 elements:

- the 28,800 elements that keep each carrier act on the glue Z/5 by ±1;
- the 28,800 elements that exchange the carrier and its twin act on the glue by ±2, the doubling;
- every exchange has order 4, 8, 12 or 20. **No involution exchanges the carrier and its twin.**

The day's quarter d^6 (§6) is one such exchange, with $(d^6)^2 = -I$. The two carriers cannot be swapped as a mirror pair; on the bath, "above and below" is a quarter-turn on the glue.

Waves (THEOREM, layer 17).

- On the K5 edges, sound is the four longitudinal modes and light is the six transverse cycle modes.
- $d_0 d_0^T + \mathrm{Curl}^2 = 5I$, so the two are isochronous at $\sqrt{5}$. Agni $= 5I - G = \mathrm{Curl}^2$ is the light operator.
- The two light helicities are the icosahedral 3 and 3′ of A5.
- No S5- or A5-symmetric medium converts sound into light.

5. The circle and the tick bit

One lemma (THEOREM, layer 19). Z/4320 has one subgroup ⟨d⟩ for each divisor d of 4320, of order 4320/d. Containment is divisibility of the gcds, and meets and joins are lcms and gcds. Two consequences:

- **Step and order are dual.** The atlas is the step 36, of order 120. The moon is the step 120, of order 36.
- **The spine is a chain** through the Coxeter subgroup:

  0 ⊂ ⟨864⟩ (tick, order 5) ⊂ ⟨432⟩ (yuga, 10) ⊂ ⟨144⟩ (Coxeter, 30) ⊂ ⟨36⟩ (atlas, 120) ⊂ Z/4320.

  The tower is therefore 36 × 4 × 3 × 2 × 5, and every element has a unique address.

By the Chinese remainder theorem, Z/4320 = Z/32 × Z/27 × Z/5, with orthogonal idempotents 3105, 2080 and 3456. The Coxeter step splits as 144 ≡ 2160 + 1440 + 864: half-turn plus sun plus tick.

The tick bit (THEOREM, layer 21). For every x, $864 \in \langle x \rangle \iff 5 \nmid x \iff 3456 \cdot x \not\equiv 0 \pmod{4320}$. Consequences:

- exactly 24 of the 48 subgroups hold the tick (the ⟨g⟩ with g | 864);
- a subgroup holds all five tick levels {0, 864, 1728, 2592, 3456} or only 0;
- every width x prime to 5 that divides 4320 satisfies x × index = 864;
- 16 of the 24 have order divisible by 20;
- the bit is not additive: 2 and 3 hold the tick, and 2 + 3 = 5 does not.

The ledger is the bit (THEOREM). The sibling's ledger holds 129 tick sentences from 35 joins:

- energy, telescopes, golden, observer loop, mirrors, W0 reverse, phase, charge, spin;
- heat, monopole, field, Einstein line, two 36s, sign tensor, Om, karma, japa, mantra, bubbles, horizon;
- above and below, tick bounds, gravity dots, K5 chain, levels, pada, SM time, cakra, smear, readout, alpha, Hubble, height.

Every one of them is the residue of a number mod 5. Their step counts (for example 27 × 192 ≡ 864) and indices (for example 864 = 27 × gcd(224, 4320)) agree, as do 38 stated subgroup orders. The readings riding on the bit are counts (ARITHMETIC), for example:

- 6 + 15 + 15 = 21 + 15 = 36, with no 18 + 18;
- the wings 21 + 7 + 77 = 105;
- the phase weights 98 + 62 = 160;
- 30/36 = 5/6;
- the monopole chain ⟨120⟩ ⊂ ⟨60⟩ ⊂ ⟨30⟩ ⊂ ⟨10⟩, none of which holds a tick.

No map joins these numbers to the objects they name.

6. One group for every clock

The Coxeter clock (SELECTED_REFERENCE; its properties are THEOREMs, layer 19). Let $c = s_1 \cdots s_8$ be the product of the simple reflections of E8. It has order 30, characteristic polynomial Φ_{30}, and exponents equal to the units mod 30. It permutes the 240 roots in eight free orbits of 30, so it is a symmetry of the bath already in the model. The selected conjugate $c' = w\, c\, w^{-1}$ satisfies

> **$c'^6 = r\, r'$**,

the K5 vertex rotation r on the carrier A4 together with the twin rotation r′ on A4′: the tick turns two carriers. The other prime-order powers are:

- the sun c'^{10}, which turns four triangles ($A2^4$);
- the half-turn $c'^{15} = -I$, which flips eight edges ($A1^8$).

Their product is $c' = (-I)\cdot c'^{10}\cdot c'^6$, which matches $144 \equiv 2160 + 1440 + 864$.

Fixed momenta (THEOREM, layers 19 and 20). On the Brillouin torus, c^k fixes $|\det(c^k - I)| = \Phi_n(1)^{8/\varphi(n)}$ momenta, where $n = 30/\gcd(k, 30)$. That gives 25, 81 and 256 at orders 5, 3 and 2, and 1 otherwise. By the gradient lemma these are critical points: the origin and 24 points of A4+A4, the origin and 80 of E6+A2, and all 256 half-periods.

The normalizer (THEOREM, layer 19). The normalizer N of $\langle c' \rangle$ in W(E8) has order $240 = 30 \times 8$. Every unit m mod 30 is realized as a twist $w\, c'\, w^{-1} = c'^m$, and the centralizer is $\langle c' \rangle$ itself. The circle's automorphisms of its Coxeter subgroup are therefore all realized on the bath. The table lists the named elements; each is in `data/clocks.json`.

element	circle	twist m	order	on the carrier
c′ (Coxeter clock)	step 144	1	30	mixes A4 and A4′
$c'^6 = r\, r'$ (tick)	step 864	1	5	rotates the pentagon on each carrier
d (day clock)	step 180	7	24	$d^4 = c'^5$; d^6 exchanges A4 and A4′
v (golden clock)	step 216	11	20	v^4 = the tick
m (mirror)	$x \mapsto -x$	$29 = -1$	2	the pentagon reflection, three atoms fixed

The tick is a commutator (THEOREM, layer 21). For the day clock d (order 24, characteristic polynomial Φ_{24}, $d^4 = c'^5$),

> $d\, c'\, d^{-1}\, c'^{-1} = c'^6 = r\, r'$.

The twist by 7 that the integrated build stops at and the two-carrier tick are therefore one fact. Write $G = \langle c', d \rangle$. Then:

- G has order 120, not the 720 of the abstract product;
- its centre is $\langle c'^5 \rangle$ of order 6, the circle's shared subgroup $\langle 720 \rangle$;
- its derived group is the tick, of order 5, and its abelianization is cyclic of order 24, generated by d;
- the day acts on the tick by doubling: $d\,(r\, r')\, d^{-1} = (r\, r')^2$.

On the circle (THEOREM, layer 22). This tick lives in W(E8). G has exactly 24 homomorphisms to Z/4320, and each one kills c'^6 and sends c′ into $\langle 720 \rangle$. The tick reaches 864

only through ⟨c′⟩ alone, by $c'^k \mapsto 144k$, and that embedding does not extend to G.

Five day clocks (THEOREM, layer 21). G has exactly five conjugates of ⟨d⟩. On them:

- the tick acts as $x \mapsto x + 1$;
- the day acts as $x \mapsto 2x$;
- the centre acts trivially.

So G modulo its centre is the affine group AGL(1, 5), the twenty affine maps of the pentagon. In it the rotation is the tick, and the doubling, which sends the pentagon to the pentagram, is the day. The ledger's statements "doubling and rotation give 20 affine maps" and "the stored doubling does not commute with the order-5 rotation" are this group, realized on the bath.

The golden clock (THEOREM, layer 21). The 24 elements of order 20 in N all have twist 11. The one with $v^2 = c'^3$ satisfies $v^4 = c'^6 = r\,r'$, so every fourth golden step is the tick. Three cyclic groups of order 20 obey the same law:

- ⟨216⟩ on the circle, where $4 \times 216 = 864$;
- ⟨v⟩ on the bath;
- $B = [[1,1],[1,0]]$ modulo 5, where B has order 20 (the Pisano period) and B^4 is unipotent of order 5.

The affine 20 of the five day clocks is not cyclic. The two kinds of 20 are different groups.

The mirror (THEOREM, layer 21). The circle's negation $x \mapsto -x$ fixes only 0 and 2160. It swaps $864 \leftrightarrow 3456$, $1728 \leftrightarrow 2592$, and the golden steps $1080 \leftrightarrow 3240$. It is realized by the 30 elements of N with twist -1, each of which inverts the clock, the tick and the sun. Ten of them are involutions that keep the carrier. On the carrier five act as a pentagon reflection, fixing one vertex and two edges, and five as its negative. The ledger's "graph reflection" and "negation on the circle" are one element of W(E8).

What can act on the bath (THEOREM, layer 19). A subgroup of the circle acts faithfully on the bath lattice exactly when its order is one of

> 1, 2, 3, 4, 5, 6, 8, 9, 10, 12, 15, 18, 20, 24, 30.

This follows from explicit elements, the degree bound, a Sylow argument for 16, and the automorphism groups of A2 (for 36) and A4 (for 40 and 60). So:

- the maximal clocks are the Coxeter clock (30), the day (24), the golden clock (20) and ⟨240⟩ (18);
- the atlas, moon, nakshatra, pada and Human Design gate cannot act. On the circle the day and the Coxeter clock commute and generate the atlas, but on the bath no element of order 24 commutes with c′.

7. The canon and the sibling, reconciled

Position. The Quick Access workbook has a new top sheet, 00_FULL_RECONCILIATION_20261028, with review cutoff 17:19:41 UTC. It keeps OMSOON736 as the controlling numbered baseline. No new Matrix ID was issued, and every frozen gate stays. It still lists this repository's 14:37 build (18 layers, 320 checks) as the latest One Machine status. At 19:10 UTC a continuation sheet, 00_CURRENT_20261028_1910, recorded the 21-layer build (412 checks, 31 tests) as source-reported. It replays the six-port moments 1, 242, 58691, 14259954, keeps OMSOON736 and every frozen gate, and still lists the order-16 coefficients as missing. It also makes two remarks, on the commutator and on the energy splits. Layer 22 answers both (§8).

The Canon Check (layer 20).

- Its preparation identities are replayed: the Bernstein ellipse, $\|C\|^2 = 495$ and the optimal $\eta^* = 2\,\mathrm{arcosh}(d/8T)$. The horizon laws $\varepsilon^{-1/2}$ and $\varepsilon^{-1/3}$ stay CONDITIONAL on the inherited exterior contract.
- Its scalar bound $(r+1)(d+2) \geq 195$ follows from layer 14.

What the sibling adopted. At 17:06 the sibling's CANON_READ adopted the corrections of layer 20:

- "360000 is not a step" is removed;
- both harmonic quartics are named with their Laplacians;
- 5 and 25 are dual to the tick;
- the fixed determinants 25, 81 and 256 are stated;
- the day–Coxeter extension has order 120.

Its 18:17 full edition keeps these, and twice notes that "the order-24 element was not constructed". Layer 21 constructs it, and `data/clocks.json` and `ports/ts/ledgerJoins.ts` carry it into the sibling's app.

The separate model, replayed (layer 20). "Finite Spectral Dynamics with Complete Observation and Exact Cyclic Addresses" is rebuilt from its definitions:

- the observer completion and the sensor census (§2);
- the 25-coordinate flag geometry, with intersection determinant -48 and vertex distance exactly $\sqrt{(8/7)}$;
- the collocated transfer functions $G_V = (z^2+7)/((z^2+3)(z^2+8))$ and $G_E = 1/(2z^2) + 2/(5(z^2+3)) + 1/(10(z^2+8))$.

8. The evening notes (layer 22)

At 19:21 UTC Drive held 28 new sibling notes (18:32–19:21) and the canon continuation of 19:10. At 19:57 it held thirteen more notes (19:22–19:54) and the separate model's status, at 20:12 three more, at 20:25 four more, at 20:33 one more and at 21:05 fifteen more. Layer 22 replays all of them in 50 checks.

Counts on the bit (THEOREM). Most of the notes read one more stored integer on the bit of §5. There are 486 readings from 63 notes, and each is checked together with what its note states: an order, a subgroup, a step count, a residue, or a first landing. The first landing is the least k for which kx is a nonzero tick. None fails.

- Twelve of the fifteen stages meet the tick, and Mātṛkā 0, Nyāsa 5 and Japa 10 do not.
- Buddhi and Pratyabhijñā generate one subgroup and take 432 steps each, but land on 864 and 1728.

Graphs (THEOREM).

- **K5.** The cut has rank 4 and the cycles rank 6. The signless kernel on the edges, of dimension 5, is the zero-mode space of D^2. The six star-tree cycles have Gram determinant 125, which is the spanning-tree count.
- **Petersen.** $(A - 3)(A - 1)(A + 2) = 0$, with exact orthogonal projectors of ranks 1, 5 and 4.
- **Pentagon and pentagram.** $C5^2 = 2I +$ pentagram. The doubling $1 \to 2 \to 4 \to 3$ carries the pentagon onto the pentagram.
- **Companions.** The breath companions of 3 and 8 have determinants 3 and 8, so no power is I. The golden companion has infinite order over $\mathbb{Z}$ and order 20 mod 5.

The root graph and the moments (THEOREM).

- **Root walks.** The closed walks at a root are 1, 0, 56, 1512 and 62160. They come from the A^2 row 56, 27, 12, 1, 0 at inner products 2, 1, 0, −1, −2.
- **The reported moments.** They are $\langle p, \Delta^n p \rangle$ for the alternating hexagon port, the six roots of an A2 with signs + − + − + −, on the infinite E8 lattice:

  > 1, 242, 58691, 14259954, 3470101592, 845582723240, 206295748792515, and μ_7 = 50383499822338790.

  All eight are computed exactly from the 240 roots, and the eighth is new. The variance is 127, the central third moment is −4736, and K_R = [[242, 127], [127, 25998]].
- **Square and octagon ports.** The canon's square-port moments $m_0 \ldots m_7$ and the octagon's 58713, 14274162 and 3476302936 are replayed. square = (hexagon + octagon)/2 at every order, because the three difference distributions are proportional on Weyl orbits.
- **The deep hole.** The deep hole is the band vertex of §3: 256 ∈ spec Δ. The full bound sup spec Δ = 256 stays reported.

The order-120 clock.

- **Structure (THEOREM).** Its element orders are 1:1, 2:1, 3:2, 4:10, 5:4, 6:2, 8:20, 10:4, 12:20, 15:8, 24:40 and 30:8. Its only involution is the central −I.
- **The Hurwitz value 6 (OBSTRUCTION).** The value 6 = 120·(−2) + 60 + 90 + 96 is 2g − 2 for signature (2, 4, 5), so g = 4. It belongs to S5, whose 120 (2, 4, 5) pairs all generate it. The stored clock has no such generating pair.

Two answers to the canon (THEOREM).

- **The commutator.** Every map from the clock to the circle kills the commutator (§6).
- **The energy.** The three splits are one conserved table, by block and D^2 sector:

	sector 0	sector 3	sector 8	total
vertex	0	3	20	23
edge	2	9	0	11
total	2	12	20	34

  The rows are 23 + 11, the columns are the stored 2 + 12 + 20, and the nonzero cells are 20 + 3 + 9 + 2.

One correction (THEOREM). FOUR_NAMES says the carrier involution does not act on the order-15 clock. By Lagrange it cannot lie in that clock, but it does act on it, by $x \mapsto -x$.

The 19:57 batch (THEOREM).

- **The six-site chain.** The six-site blocks 241I − J, 183I + 52J and 35366I + 8134J are the block Stieltjes data of six mutually adjacent lattice points (an A5 root simplex): M_1, $M_2 - M_1^2$ and $M_3 - 2M_1M_2 + M_1^3$. The canon listed the third as unidentified. Its uniform values are 235, 495 and 84170.
- **Beats.** The separate model's squared beat frequencies 11 ± 4√6 are $(2\sqrt{2} \pm \sqrt{3})^2$, the beats of the carrier's two oscillating sectors.
- **SEVEN_ONE, corrected.** Four mirror pairs k, 14 − k, not three, are left unsplit by the bit: 1–13, 2–12, 3–11 and 6–8.

- **Rebuilt.** The oriented operator (spectrum 0^7, 5^8), the golden family (B^7 = [[21, 13], [13, 8]]) and Agni = Soma = $I - B^TB/5$.

The 20:12 notes (THEOREM).

- **The Petersen projectors.** The sibling's projectors $J/10$, $(4I + 2A - J)/6$ and $(5I - 5A + J)/15$ have denominators 10, 6 and 15 and add to I over 30. The Petersen graph is the complement of the line graph of K5, so $A = J + I - M^TM$ commutes with the edge stiffness, and the projectors are the carrier's own: onto sector 8, the five dark modes and sector 3. The denominators are the gap products 2·5, 2·3 and 3·5, and only 6 avoids the spread 5. The Gram determinant 648 is det(3I + J), the carrier's vertex block.
- **Every breath.** The breaths of 3 and 8 share no period because 8/3 is not a rational square. "Not a multiple" is not the criterion: 4 and 9 share a period.

The 20:25 notes (THEOREM).

- **The Pauli generator is the bath.** The sibling's saved generator $L(M) = 240M + 16M^T - 16\,(\mathrm{tr}\,M)\,I$ on real 16 × 16 matrices has eigenvalues 0, 224 and 256 on 1 + 120 + 135 Pauli directions. Eight antisymmetric 4-qubit Paulis anticommute exactly along the E8 Dynkin diagram, so their products give a bijection φ from the 256 half-periods $\Lambda/2\Lambda$ onto the Paulis, with $L\,\varphi(x) = \varepsilon(\pi x)\,\varphi(x)$. So L is the bath Laplacian on its half-period modes: 0 at the origin, 224 at the 120 E7+A1 saddles (the canon's 224) and 256 at the 135 D8 band-top points.
- **Mass and energy.** The "mass determinant" 216 is det(3W) for the source metric W of layer 10 (det W = 8/9). The datum's "2" is the kinetic energy of a zero-stiffness 4-cycle in ker M, not a frequency.
- **Written out.** `data/pauli_half_periods.json` lists the eight Dynkin words (IIIY, IIXY, IIYX, IYZI, XIXY, ZXXY, IZXY, IXXY) and all 256 half-periods with their words and band values. The layer rechecks it from the file alone, and `ports/ts/pauliBath.ts` replays it in the sibling's app, where it had been recorded as not replayed.

The 21:05 notes (THEOREM).

- **The rotor plane.** The C5 rotation fixes exactly one plane of the carrier's cycle space, spanned by the oriented pentagon and pentagram, and the doubling turns it a quarter (KARMA_K2, OM_ROTOR).
- **The tick doubling.** On the ticks the doubling equals $x \mapsto 7x$, a bijection of the whole circle and the day's twist, so the rotor law is the restriction of an invertible map.
- **The bath's spectrum.** The half-period modes are 2Λ-periodic eigenfunctions of the bath: an invariant subspace of all functions on Λ, not of ℓ^2, with eigenvalues inside the band. This answers PAULI_BLIND.
- **Rotor chain.** ROTOR_CHAIN's shell tuple (1, −2, 0, 2, −1) is the hexagon's difference distribution. Its "order 24 meets only 0" is corrected: the step 180 misses, and the order 24 meets.

Not replayed (REPORTED), not formed (UNFORMED).

- **Not stored here:** the sibling's permutations on the fifteen indices, its 24-cycle readers, its Hodge degrees and heat spectrum, its 12 × 12 sense stiffness, and the acoustic and optical matrices.
- **Not formed:** Yukawa blocks, taste, smell and touch matrices, an icosahedral adjacency, the named figures and a qubit. No operand is stored for any of them.

9. The unified ledger

`docs/LEDGER.md` lists every join of both builds and of the evening notes, with its status and the layer that checks it. A test keeps it current. In short:

join	unified statement	status
circle lemma, tower	subgroups = divisors; 36 × 4 × 3 × 2 × 5	THEOREM
all tick joins	one bit: 5 ∤ X	THEOREM (the readings are ARITHMETIC)
rotor	$c'^{6} = r\ r' = [d, c']$	THEOREM
golden subgroup	⟨216⟩, ⟨v⟩, B mod 5: the fourth power is the tick	THEOREM
oscillating tick	the day doubles its five conjugates; G/Z = AGL(1,5)	THEOREM
mirrors, W0 reverse	twist −1 involutions; the pentagon reflection on the carrier	THEOREM
above and below	no involutive exchange of carrier and twin; $i \mapsto 14 - i$ does not keep D^2	THEOREM
energy split	one conserved table: vertex (0, 3, 20), edge (2, 9, 0); the stored 2 + 12 + 20 are its columns	THEOREM
observer	five channels; RH = I_{15}	THEOREM
spin counts	240 = D8 + one half-spin orbit; 135 = the band-top D8 family	THEOREM
field	$\ddot{q} + 4q = 0$ is not a carrier mode	OBSTRUCTION
sign tensor, entanglement	10 + 10 is a count; the carrier cannot entangle V with E	CONDITIONAL
charge, monopole, Einstein line, two 36s, Om, karma, bubbles, horizon	counts riding on the bit	ARITHMETIC
order-24 element	constructed, in `data/clocks.json`	THEOREM
evening counts (28 notes)	223 readings, all the bit, with orders and first landings	THEOREM
reported moments	the hexagon port on the E8 lattice; μ_7 new	THEOREM
square and octagon ports	square = (hexagon + octagon)/2 at every order	THEOREM
commutator and 864	every map to the circle kills it	THEOREM
Hurwitz value 6	S5 (genus 4), not the stored clock	OBSTRUCTION
six-site moment chain	M_1, $M_2 - M_1^2$, $M_3 - 2M_1M_2 + M_1^3$ of the A5-simplex port	THEOREM
seven, one, seven	four mirror pairs unsplit by the bit, not three	THEOREM
Petersen projectors	the carrier's edge projectors: $A = J + I - M^{T}M$, the complement of the line graph	THEOREM

join	unified statement	status
Pauli generator	the bath Laplacian on its 256 half-period modes (2Λ-periodic, not ℓ^2)	THEOREM
rotor	the C5-fixed plane of the cycle space, turned a quarter; the tick doubling is the day's twist ×7	THEOREM
source infinity, foam, double slit, photoelectric, electron, photon, proton, Compton, Bohr, Bell count, 19 SM parameters	no operand, no dimensionful constant	UNFORMED

10. What is not formed, and the gates

Not formed.

- The six SCR6 operands, the original seams, the 502- and 237-dimensional operators, the MPKG adapter, and the historical Agni, Soma, Suraj, Sawan and Om matrices.
- The Krishna lift's operands, the epoch and the ayanamsha.
- GKSL, SI units, BRST, BV and Einstein–Hilbert.

The carrier action is constant-free, so it can predict only ratios such as √(8/3). A length or a mass needs one authenticated dimensionful datum, which is TIME_BIND.

Gate	State	What closes it
AUTH / SCR6	NOT_PASS / operand absent	the owner's signature; the six SCR6 operands of a chosen physical child
TIME_BIND	UNBOUND	one authenticated clock datum (frequency ratios are calibration-invariant)
DYN_BIND	NOT_INSTALLED	the physical child, the generation multiplicity, the chirality index
PredictionFreeze	NOT_TRIGGERED	the owner freezes the register before new comparisons
W_L	NOT_PASS by evidence	the recorded PMNS candidate is excluded ($\chi^2 = 10.6$ on 2 dof)
T3081	UNOPENED	the official host must be allowed
ScientificGold	FALSE	a frozen prediction, independent data and a preregistered test

Open mathematics.

- The convexity collar (223.58, 224).
- The rational coefficients of L_{16}, and a proof that it annihilates W_n for every n.
- A preparation protocol matching the $\varepsilon^{-1/2}$ and $\varepsilon^{-1/3}$ bounds.
- The eight-dimensional dispersive constant.

Appendix A. The twenty-two layers

#	layer	checks	content
1	carrier	36	K5, D, spectrum, Lagrangian, exact motion, Petersen, Hodge block, edge tick, harmonic quartics
2	spectral	9	heat trace, ζ_D, spectral dimension, moments

#	layer	checks	content
3	spacetime	34	RP^2 pair, clock 125 → 120, E8 roots, walks, symbol, Green, clique
4	native	19	source, port and bath action; W_AS; 40/9; radiation channel; nonlinear flow
5	observer	7	readouts, dark sector, Witten index −5
6	gravity	8	Lin–Lu–Yau curvature; spectral action (CONDITIONAL)
7	cosmology	7	FLRW with the carrier field (CONDITIONAL)
8	frontier	23	band-edge orbit 135, the 224 saddle, saddle law, collar survey, coil
9	joins	24	the maps that make the layers one object
10	literal	20	the recovered operand file, every identity replayed
11	glue	11	E8 = D8 + s; A4 ⊕ A4′ with x ↦ ±2x
12	binding	12	DYN_BIND constraints
13	gates	12	TIME_BIND courts, register, W_L, T3081, AUTH package
14	recurrence	16	W_{90} exact, W_{600} mod p, exclusion, the order-16 operator
15	court	19	Court U19.81 audited
16	handover	30	the Locked Memory handover audited; critical census; T_6|V_c
17	waves	18	sound, light, frequency and phase on the K5 edges
18	snapshot	16	the 14:16 snapshot rebuilt; T^9 density; Hessians; asymptotics
19	spine	50	the circle lemma, the Coxeter clock, two carriers, the 7, what can act
20	canon	22	the canon read at 16:30 and 16:49; √30; the separate model replayed
21	ledger	19	the tick bit; the clock group; the five day clocks; golden; mirrors; carrier exchange; energy; spin; the clock file
22	counts	50	the evening notes: 486 readings on the bit; K5, Petersen and its projectors, pentagon; root walks; the hexagon, square, octagon and six-site ports; the 120-clock; circle maps; the energy table; beats; mirror pairs; breaths; the Pauli generator as the bath, written out; the rotor plane; two Laplacians and the line graph

Appendix B. Replay

```
pip install -r requirements.txt
python verify.py              # 22 layers, 462 checks, about 4 minutes
python -m pytest -q           # 39 tests
node --experimental-strip-types --no-warnings ports/ts/run.mjs   # the three TypeScript audits
```

Sources: the Drive folder "One Machine 20261028" (the sibling's full edition, `FULL_UPDATE_20261028.md` and 104 notes of 17:01–21:03 UTC) and the canon sheets 00_FULL_RECONCILIATION_20261028 and 00_CURRENT_20261028_1910.